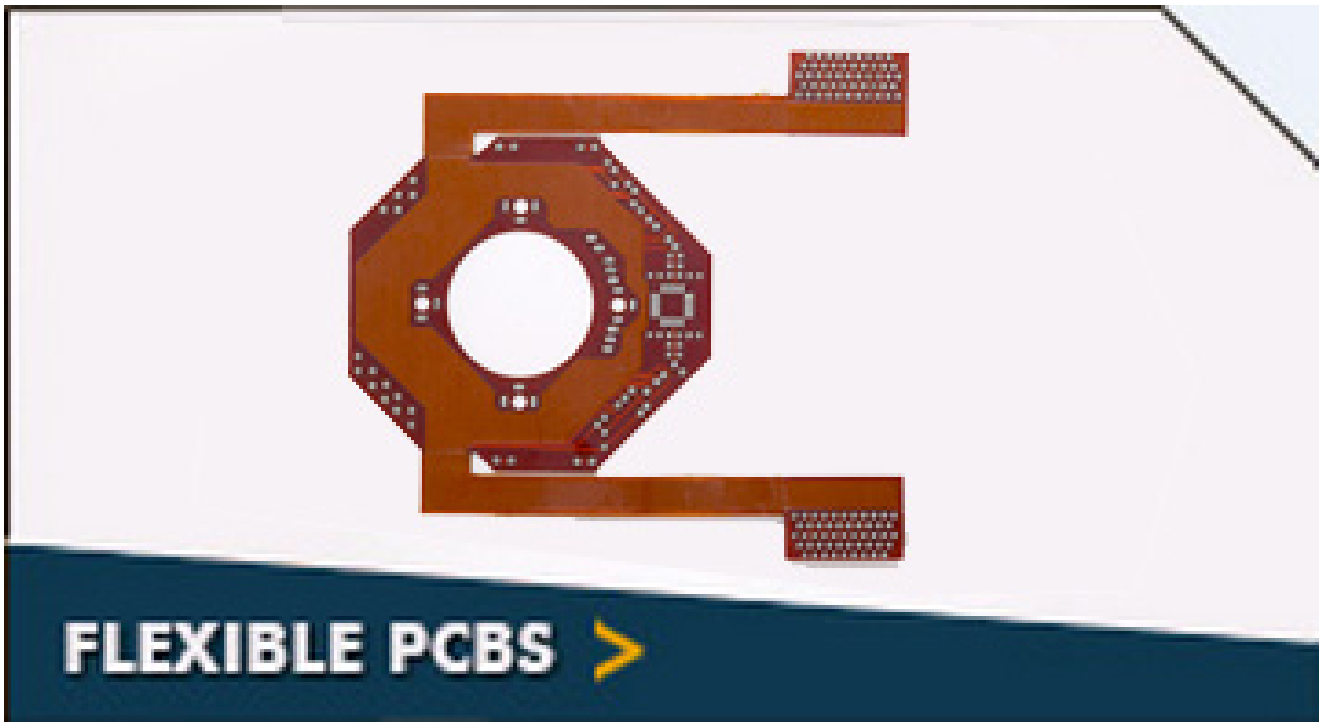
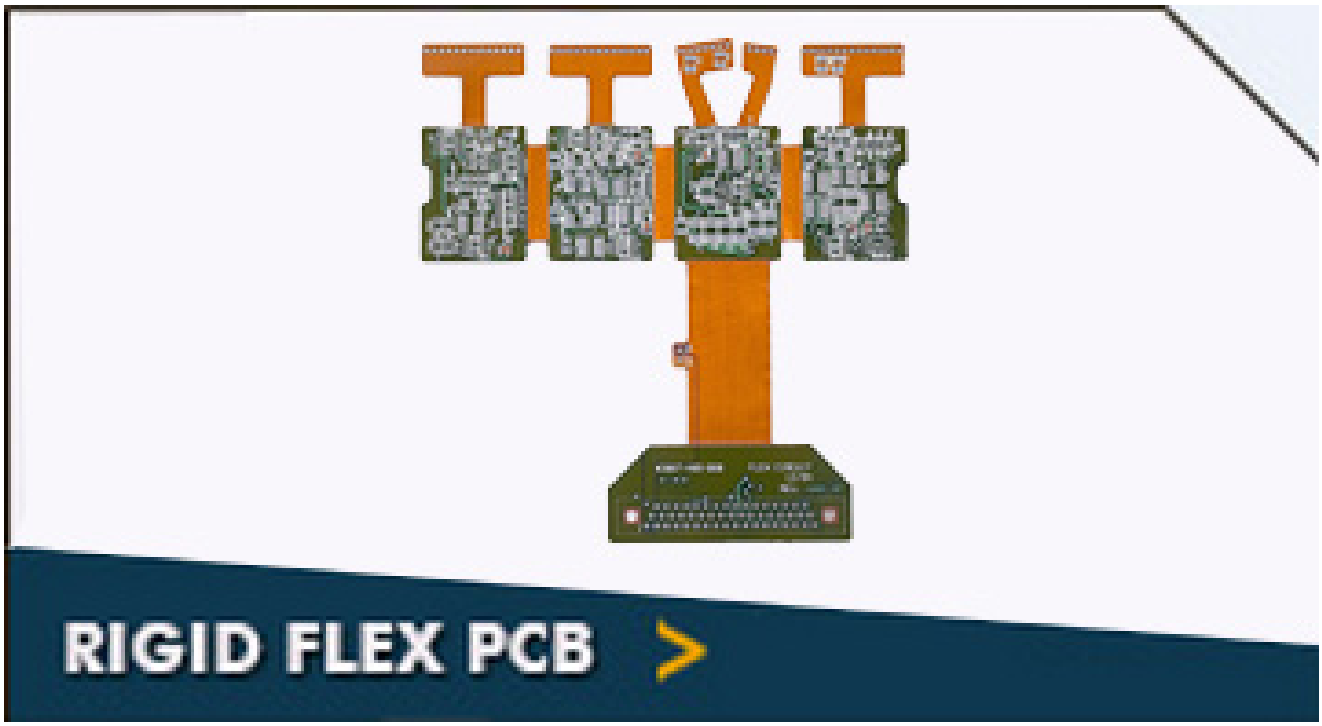
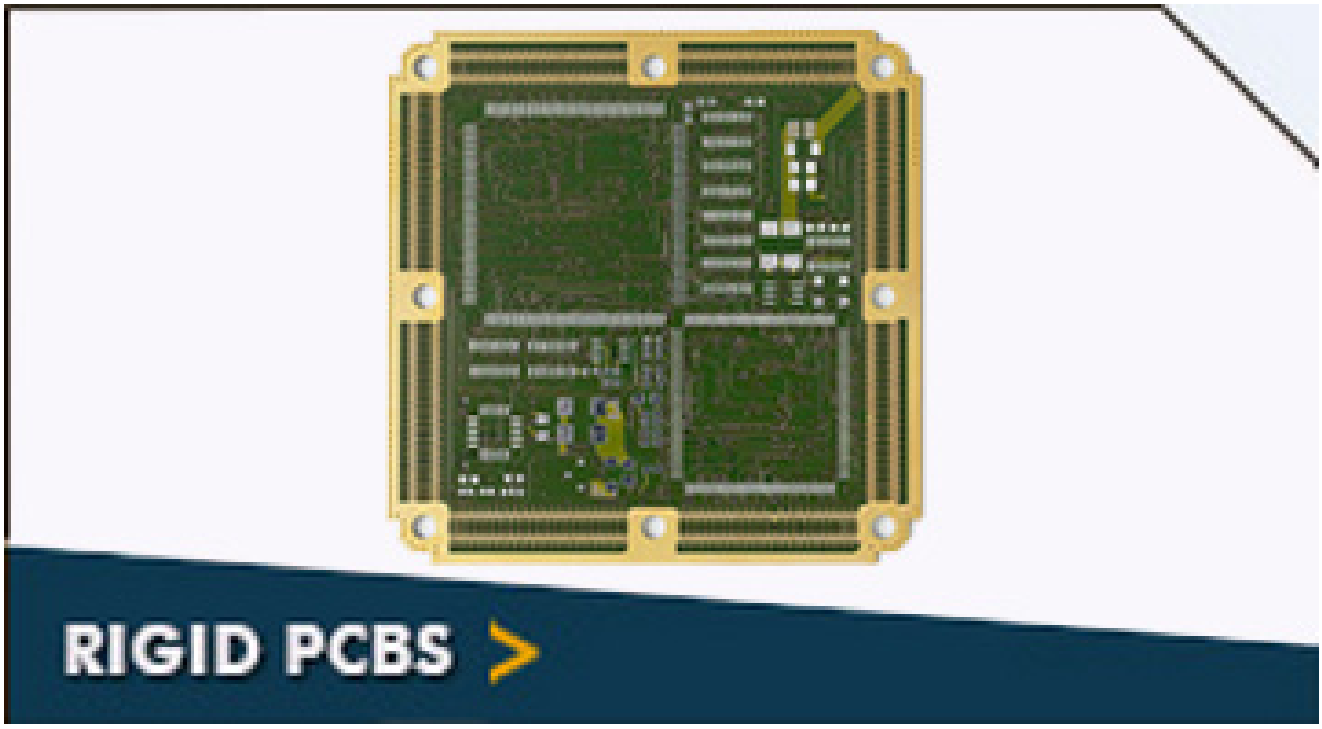
RIGIFLEX TECHNOLOGY, INC. (/)
Manufacturers of Rigid, Flex and Rigid-Flex Circuit Boards
(714) 688-1500 (tel:7146881500) pcboards@rigiflex.com
(mailto:pcboards@rigiflex.com)
Flex & Rigid-Flex PCBs.. 1-5 Day Quick Turn



(https://www.rigiflex.com/products-services/flexible-pcb/)



(https://www.rigiflex.com/products-services/rigid-flex-pcb/)



(https://www.rigiflex.com/products-services/rigid-pcb/)

Superior & Trusted Flexible, Rigid-Flex & Rigid PCBs for Over 25 years

A Detailed Description of the Printed Circuit Board Fabrication & Manufacturing Process

This document provides a step by step process of the PCB manufacturing process. The process given below provides information for manufacturing a double sided printed circuit board:

- **Collecting Data from the Customer:** Data provided by the customer regarding the printed circuit board is taken into consideration. The Gerber files are used to extract the data. Plots and photo plots

are developed accordingly.

- **Creating an Output File:** A PCB software (e.g. CadSoft, Eagle, and Freeware) is used standardize the Gerber data. A processed output file in the format of a Gerber extension is created as a preview of the PCB design. Once the design of the PCB is approved, the other processes of manufacturing begin.
- **Material Arrangement:** Standard PCB panels are 1.6mm thick, with copper coatings on either side. These panels are cut down to size along with a backup board and the drill entry material.
- **PCB Drilling:** Two activities take place at this stage. First, all tooling, and reference holes are drilled into the panels. A CNC drilling machine is used to produce component and through hole plated drills.
- **Through Hole Plating:** An electrographic film is plated to the drill cavity wall. This will support ensuing stages, which will require the copper to be galvanized.
- **Resist Lamination, Exposure, and Development:** There are three steps in this stage:
    - First, the PCB panel is laminated with dry resist film using high temperature and pressure.
    - The resist is then exposed to ultra violet rays by using the photo plots.
    - The panel is immersed in a solution made from sodium carbonate to develop the panel. Once this has been completed, the electroplating process can begin.
- **Electroplating:** The PCB's pads and tracks are electroplated with copper, along with tin film. The film acts as a protective coating, and protects the components during the etching process.
- **Resist Stripping:** A corrosive solution is used to strip away the photoresist. This is performed to support the forthcoming immersion and spray coat processes.
- **Etching:** The etching process consists of two stages:
    - An ammonia solution coat is sprayed over the copper film to remove any excess material. The galvanized tin protects the components.
    - The board is placed in a rinse tank for a few seconds, removed, and then dried.
- **Tin Film Removal:** Only when the etching process has been completed can the protective tin film be removed. This is performed by using an acid based tin stripper like nitric acid. The tin stripper phase supports PCB spraying processes.
- **Soldermask:** The soldermask stage consists of three processes:
    - **Application:** There are two ways the soldermask can be applied – either as a liquid varnish, or as a dry film. The soldermask is applied using a hanging casting implement. Once this has been completed, the spraying and screen printing processes can take place.
    - **Exposure:** Like the resist, the soldermask is also exposed to light using the photo plots.
    - **Development:** The PCB is then dipped in a sodium solution. This ensures that any excess soldermask is removed from the pads and soldering points.
- **Silkscreen Printing:** Once the soldermask processes have been completed, a silk screen is immediately printed on the mask. This process is performed using a legend printer. The printer applies the screen print directly onto the PCB.
- **Soldermask Tempering:** The PCB is placed in an oven at a temperature of 150°C for 60 minutes. This helps temper the soldermask.
- **Surface Finishing:** For the surface finishing process, an ENIG surface plating is being used here. As a surface finish, ENIG provides benefits of a flat surface, and acts as a stress free coating. ENIG is a combination of chemicals, nickel, and gold. Using vertical baths, the pads are surface coated with ENIG. The nickel surface ensures solderability, while the gold layer protects the nickel.
- **PCB Routing:** The final stage is that of PCB routing. However, many times, the panel moves during this phase. To prevent this from taking place, the panel is first pinned down onto the routing machine bed. The PCB is routed using a CNC routing machine. The machine consists of a spindle that moves at fast speeds, providing a high feed rate.

Once all these processes have been completed, the PCB is tested and then sent for client delivery. Knowledge of all the stages of PCB manufacturing (https://rigiflex.com/capabilities) can help clients make better decisions, and understand the difficulties faced by PCB manufacturers.

PCB Manufacturing Capabilities		
Specifications	**Standard Technology**	**Advanced Technology**
Number of Layers	1- 12	14 – 40
Board Material	FR4 FR406 IS410 Kapton Shin-Etsu – Epoxy Adhesive System Rogers – Epoxy Adhesive System Dupont FR – FR Acrylic Adhesive System Dupont LF -LF Acrylic Adhesive System Adhesiveless Base Materials	Aluminum Core Arlon CEM Copper Core FR408 Getek Nelco 4000 Rogers 3000 Rogers 4000 Rogers 5000 Taconic TLY Polyimid
Minimum Board Thickness	2 layer – 0.010″ 4 layer – 0.020″ 6 layer – 0.020″ 8 layer – 0.062″ 10 layer – 0.062″ 12 layer – 0.062″	2 layer – 0.005″ 4 layer – 0.010″ 6 layer – 0.031″ 8 layer – 0.040″
Maximum Board Thickness	2 layer – 0.125″ 3-12 layer – 0.200″	0.250″
Maximum Board Size	16″ x 22″ 12″ x 21″ 22″ x 28″	10″ x 16″ 16″ x 22″ 12″ x 21″ 22″ x 28″
Copper Thickness	0.5 oz -3 oz	4oz – 6 oz
Hole Aspect Ratio	7: 1	15:01
Minimum Hole Size	0.008″	0.006″
Minimum Trace/Space	0.006″/0.006″	0.003″/0.003″
Minimum	0.010″	0.003″
Drill-to-Copper		
Minimum Pitch	1 mm	0.3 mm

Final Finish	HASL (Solder) Lead Free Solder Copper Gold Gold Fingers White Tin OSP	HASL Gold (ENIG/Hard/Soft) Selective Gold Immersion Silver OSP White Tin
Solder Mask	LPI: Green Black Red Blue Yellow White Clear	LPI: Green Black Red Blue Yellow White Clear Mix-and-match Wet Mask Dry Film
Coverlay	FR Coverlay LF Coverlay Shin-Etsu Flexible Soldermask	
Silk Screen	White Black Yellow	White Black Yellow Green Red Blue
PCB Fabrication	Scoring Route & Retain	Jump Scoring Route & Retain Milling
Additional Features	Plated Slots Non-plated Slots Controlled Dielectric Covered Vias Counter Sinks Counter Bores Dual Access Flex Suspended Leads	Plated Edges Plated Milling Plated Counter Bores & Counter Sinks Edge Castellation Controlled Impedance Silver Filled Vias Non-Conductive Filled Vias
Quality Standards	IPC 6012 Class 2 Electrical Testing 100% Netlist Testing TDR Testing	Milspec 31032 Milspec 55110 Milspec 50884 IPC 6012 Class 3 100% Netlist Testing TDR Testing

Special Technology		Blind & Buried Vias Laser Drilled Vias Mechanically Drilled Micro Vias Metal Core Boards Burn-in Boards Rigid-Flex Boards Flex Boards Micro Circuits

Quick Contact

All Fields are Mandatory (*)

Full Name

Email

Telephone

Type Your Message

ed3c23

Captcha

Submit



Need it quick? That is our speciality!

FOOTER NAVIGATION

» Home (https://rigiflex.com/)

» Company Profile (https://www.rigiflex.com/company-profile)

» Customers (https://www.rigiflex.com/company-profile/customers)
» Capabilities (https://www.rigiflex.com/capabilities)
» Product & Services (https://www.rigiflex.com/products-services)
» Flexible PCB`s (https://www.rigiflex.com/products-services/flexible-pcb)
» Rigid-Flex PCB`s (https://www.rigiflex.com/products-services/rigid-flex-pcb)
» Rigid PCB`s (https://www.rigiflex.com/products-services/rigid-pcb)
» Quality & Delivery (https://www.rigiflex.com/quality-delivery)
» Industries Served (https://www.rigiflex.com/industry-served)
» Contact (https://www.rigiflex.com/contact-us)
» Privacy Policy (https://www.rigiflex.com/privacy-policy)
» PCB Quote (https://www.rigiflex.com/pcb-quote)
» FAQ (https://www.rigiflex.com/faq)
» Blog (https://rigiflex.com/blog/)
» Sitemap (https://www.rigiflex.com/sitemap)
» Case Study (https://www.rigiflex.com/case-study)
» Certifications (https://www.rigiflex.com/certifications

- XML (https://www.rigiflex.com/sitemap.xml)

FOLLOW US ON

Facebook
(https://www.facebook.com/RigiflexTechnologyInc)

Twitter
(https://twitter.com/rigiflextech)

Rigiflex Technology Inc.

1166 N. Grove Street Anaheim, CA 92806

(714) 688-1500 (tel:7146881500)

(714) 688-1505

pcboards@rigiflex.com (mailto:pcboards@rigiflex.com)

© 2025 RIGIFLEX TECHNOLOGY INC. - EXPERTS IN FLEX AND RIGID-FLEX CIRCUIT BOARDS MANUFACTURING